

Jason Staker

Trabuco Canyon, CA, USA | U.S. Citizen | [LinkedIn](#) | [Website](#) | 949-556-5846 | jason.staker@yahoo.com | [GitHub](#)

EXPERIENCE

Software Developer

September 2025 - Present

UBC Uncrewed Aircraft Systems (UAS)

Vancouver, BC, Canada

- Built a ROS 2 mission-control node running on a Jetson Nano that converts perception-relative target poses into MAVROS setpoints, flying stepwise with dwell timing to a commanded stand-off distance.
- Implemented mission-state gating that rejects setpoints unless the vehicle is connected, armed, in GUIDED mode, and publishing valid pose, cancelling cleanly if any condition drops mid-flight.
- Built a perception bridge node converting RealSense detections into body-frame pose commands using camera intrinsics and depth-patch sampling, with confidence gating before approach.
- Containerized a legacy C++ camera/telemetry pipeline in multi-stage Docker for ARM and x64, resolving SDK/CMake linkage failures and a fake-camera segfault; validated in SITL before field deployment.

Software Developer Co-op

September 2026 - Present

Visier Inc.

Vancouver, BC, Canada

- Platform Framework team: authentication and authorization services, framework and dependency upgrades, and internal infrastructure automation.

Software Developer (Contract)

May 2026 - Present

Staker Legal Technology

Remote

- Sole developer of a 12.5k-line Python system; enforced architectural layering with a test that parses the codebase's own imports, holding 527 tests green with strict typing and CI on Linux and Windows.
- Fuzz-tested the routing engine across 7,389 generated inputs with zero exceptions; implemented crash-safe persistence with atomic writes and advisory locking.

Software/Business Development Intern

May 2025 - September 2025

HOTB Software Solutions

Irvine, CA, USA

- Built a cross-platform Python automation platform with a PyQt5 GUI and modular region-specific scrapers, cutting manual RFP retrieval time by ~80%.

EDUCATION

University of British Columbia

August 2024 - April 2028

B.Sc. Computer Science | GPA: 4.0

Vancouver, BC, Canada

- Transport Canada RPAS Advanced License, March 2026

PERSONAL PROJECTS

Rocket Flight Simulator | [GitHub](#)

June 2026 - Present

- Building a rocket flight simulator with generic templated RK4 integrator; verified fourth-order convergence by step-halving and agreement with the closed-form solution to 1e-5 over 1,000 steps.
- Modeled propulsion from measured RASP test-stand data with impulse-proportional mass depletion.
- Implemented the seven-layer US Standard Atmosphere 1976 model matching densities to 1e-3 kg/m³.

Schwarzschild Ray Tracer | [GitHub](#)

July 2025 - August 2025

- Built a multithreaded C++17 renderer that simulates light paths around a Schwarzschild black hole using numerical integration of null geodesics.
- Used OpenMP to accelerate rendering and GoogleTest to validate core math and trajectory logic.

2D Physics Engine & Renderer Demo | [GitHub](#)

May 2025 - June 2025

- Built a real-time C++17/SFML rigid-body engine with quadtree broad-phase collision detection.
- Sustained ~1000 entities at 60 FPS and added JSON-configurable simulation parameters plus interactive debug overlays.

SKILLS

Languages: C++17, Python, Java, Bash

Embedded/Robotics: ROS 2, MAVROS, MAVLink, Jetson Nano, Raspberry Pi, ARM/x64 Linux, RealSense, OpenCV, multithreading

Numerical/Simulation: Runge-Kutta integration, ODE systems, rigid-body dynamics, Eigen, OpenMP, SITL

Testing/Tooling: GoogleTest, Catch2, pytest, CMake, Docker, Git, GitHub Actions